

Unsupervised Learning

K-Means Clustering

Unsupervised Learning

- **Clustering Algorithms**

- Example(s) of applications: *customer segmentation, image segmentation*;
- Algorithm(s): *K*-means clustering, hierarchical clustering.

- **Outlier Detection Models**

- Example(s) of applications: *fraud detection, cybersecurity, medical diagnosis*;
- Algorithm(s): DBSCAN

- **Latent Variable Models**

- Example(s) of applications: *data preprocessing, feature engineering, data visualization, data compression*;
- Algorithm(s): Principal Component Analysis (PCA)

K-Means Clustering

Assume that we have n data points, labeled $\{1, \dots, n\}$. Let $C_i, i = \overline{1, K}$ denote the clusters, i.e., each C_i is a subset of $\{1, \dots, n\}$. In addition to this, the following properties hold

- 1) $C_1 \cup \dots \cup C_K = \{1, \dots, n\}$, i.e., each label in $\{1, \dots, n\}$ belongs to at least one cluster;
- 2) $C_k \cap C_{k'} = \emptyset$, for all $k \neq k'$, i.e., the clusters are non-overlapping; no observation belongs to more than one cluster.

How to construct these clusters?

For each cluster k , look at the *within-cluster variance* $W(k)$ and minimize overall cluster variance.

K-Means Clustering

- **Within Cluster Variance** $W(k)$,

$$\begin{aligned} W(k) &= \frac{1}{|C_k|} \sum_{i,i' \in C_k} \|\mathbf{x}^{(i)} - \mathbf{x}^{(i')}\|_2^2 \\ &= \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p \left(x_j^{(i)} - x_j^{(i')} \right)^2, \end{aligned}$$

where $|C_k|$ denotes the number of observations in C_k .

- **The minimization problem,**

$$\begin{aligned} &\min_{C_1, \dots, C_K} \sum_{k=1}^K W(k) \\ &\Leftrightarrow \min_{C_1, \dots, C_K} \left\{ \sum_{k=1}^K \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p \left(x_j^{(i)} - x_j^{(i')} \right)^2 \right\} \end{aligned}$$

K-Means Clustering. The Algorithm

- 1) **Initialization step.** Randomly assign a label (from $\{1, \dots, K\}$) to each of the n observations.
- 2) **Iterate until the cluster assignments stop changing.**
 - (a) For each cluster $k \in \{1, \dots, K\}$, compute its centroid $\bar{\mathbf{x}}^{(k)}$

$$\bar{\mathbf{x}}^{(k)} = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}^{(i)} \Leftrightarrow \bar{x}_j^{(k)} = \frac{1}{|C_k|} \sum_{i \in C_k} x_j^{(i)}, j = \overline{1, p}.$$

- (b) Assign each observation to the cluster whose centroid is closest (*closest* is to be understood in terms of the Euclidean distance).

Properties of the Algorithm

The minimization problem,

$$\min_{C_1, \dots, C_K} \left\{ \sum_{k=1}^K \frac{1}{|C_k|} \sum_{i, i' \in C_k} \left\| \mathbf{x}^{(i)} - \mathbf{x}^{(i')} \right\|_2^2 \right\}$$

- The algorithm is guaranteed to decrease the value of the objective function at each iteration. Useful identity,

$$\frac{1}{|C_k|} \sum_{i, i' \in C_k} \left\| \mathbf{x}^{(i)} - \mathbf{x}^{(i')} \right\|_2^2 = 2 \sum_{i \in C_k} \left\| \mathbf{x}^{(i)} - \bar{\mathbf{x}}^{(k)} \right\|_2^2.$$

- It is not guaranteed to give the global minimum.

Optimal Value of K

